

Threshold Entry in a Fixed-Prize Contest

Proofs addendum

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The results below are proved in Lean and are more general than the statements of `MANUSCRIPT.tex`. Notation is that of Appendix A there.

Dominance with $\mu \leq 1$. $F \leq G$ holds for every $\mu \leq 1$, with no lower bound on μ , because the term μr is common to both scores.

Lean: `EntryContestP1P2.p2_fosd`.

Saturation for every incumbent. $\Delta(m) \leq V/(m+1)$ for every Q needs no atoms in the investor's law only. The non-investor's law and the incumbent's law C can be any probability laws.

Lean: `EntryContestP6P7.p7_Delta_le, EntryContestP6P7.p7_cap`.

The benchmark limit from both sides. $\Delta(m) \rightarrow V/(m+2)$ as $\mu \rightarrow 0$ in a two-sided neighbourhood of 0, for any law of r , when $s \geq 0$ has no atoms.

Lean: `EntryContestP6P7.p6_limit`.

Strictness under a spread with weaker hypotheses. The strict fall of k^* under a strong enough linear spread needs only a non-increasing κ above c , the divergence of κ at c and a non-increasing Δ . Neither the continuity of κ nor the strictness of burden-monotonicity is used.

Lean: `EntryContestKappa.p9_strict, EntryContestKappa.p9_monotone`.

The divergence primitive. For u continuous at c , $\kappa(w) \rightarrow \infty$ as $w \downarrow c$ exactly when $u(x) \rightarrow -\infty$ as $x \downarrow 0$.

Lean: `EntryContestKappa.kappa_diverges_iff`.

Burden-monotonicity without curvature. If $u'(w) < u'(w-c)$ for every $w > c$, κ is strictly decreasing on (c, ∞) . A convex stretch of u wider than c rules burden-monotonicity out, and every width below c is attained by an admissible u .

Lean: `EntryContestBM.burden_strictAnti_of_deriv, EntryContestBM.not_burden_of_convex_stretch, EntryContestBM.admissible_width`.

The no-collapse bound in general. When non-investor scores never exceed x_G , for $m < Q$

$$\Delta(m, Q) \geq V \left(\int_{x \geq x_G} C F^m dF - \frac{1}{Q-m} \right).$$

Lean: `EntryContestRefute.r1_lower_bound`.

Count invariance over all indices. The count-invariance argument holds with the outsider condition imposed at every index, not only below Q .

Lean: `EntryContestMM.count_invariance`.